AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Final Examination
Fall Semester 2025/2026

Date: 19 /Jan./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART – ONE Total Marks 10

Question No 1: Select the correct choice for each of the following statements.

1. Which component in a CNN helps reduce the spatial dimensions of feature maps while retaining key information?

A) Convolution layer
B) Pooling layer
C) Fully connected layer
D) ReLU activation

2. In an RNN, what allows the network to retain information from previous time steps?

A) Convolution filters
B) Recurrent connections
C) Dropout layers
D) Batch normalization

3. Which of the following NLP techniques is primarily used to convert text into numerical vectors?

A) Lemmatization
B) Stemming
C) Word embedding
D) Tokenization

4. What is a common problem in training deep RNNs?

A) Overfitting due to too few layers
B) Vanishing or exploding gradients
C) Lack of nonlinear activation
D) Insufficient training data

5. In a content-based recommender system, recommendations are generated based on:

A) User demographic data
B) Similarities between item features
C) Other users' preferences
D) Randomized suggestions

PART – TWO Total Marks 15

Question No 2: Differentiate between CNN and RNN architectures with examples. **(6 Marks)**

Question No 3: Explain the concept of overfitting in deep learning and mention at least one method to reduce it. **(5 Marks)**

Question No 4: What are recommender systems and give at least three examples **(4 Marks)**

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PART – THREE Total Marks 25

For the following questions, you are required to write the python code.

Scenario: Predicting Population Growth with Deep Learning

You are employed as a Lead Data Scientist at a global health analytics organization, where your team is developing an AI-powered system to predict the future spread level of an infectious disease. The goal of the project is to estimate a risk score (0 to 10) for different regions across the world, indicating the likelihood of disease spreading in the coming week.

To achieve this, you have been provided with a comprehensive dataset named "*health_surveillance.csv*", which contains 250,000 rows of real-world data. Each row represents a region-week record, capturing both environmental and health-related features that can influence disease transmission patterns. The dataset includes several types of information. Firstly, it contains satellite-derived environmental features, such as temperature patterns, rainfall intensity maps, and humidity grids, which reflect weather and climate conditions known to affect disease dynamics. Secondly, it provides population mobility heatmaps, encoded as 16×16 grayscale matrices, representing human movement intensity across regions - a key factor in understanding how infections spread geographically. In addition to these, the dataset includes healthcare capacity indicators, such as the number of ICU beds per 1,000 people, a staff shortage index, and vaccination coverage, all of which represent the readiness and resilience of each region's healthcare system. Furthermore, each record also contains historical disease intensity scores for the previous four weeks, offering temporal insight into past infection trends.

Finally, the dataset contains the target label - *future_risk* - which represents the predicted disease risk score for the following week. Your primary task is to design and train a deep learning model based on Convolutional Neural Networks (CNNs) that can effectively learn from these spatial and numerical features to predict the *future_risk* value for each region.

Question No 5: Loading the Dataset **(3 Marks)**

Write the Python code to load the dataset "health_surveillance.csv" into a "dataset" DataFrame.

Question No 6: Dataset Splitting **(5 Marks)**

Write the Python code to split the "dataset" into training and testing datasets.

Question No 7: CNN Architecture Design **(12 Marks)**

Design a CNN architecture with 3 layers for predicting *future_risk*. Justify the choice of layers, activation functions, and any other architectural decisions. You have the flexibility to adjust the number of layers as needed.

Question No 8: Model Training and Prediction **(5 Marks)**

Write a Python code to train the designed CNN model using the training dataset. Also, provide code for making predictions on the test dataset.

End of questions

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